

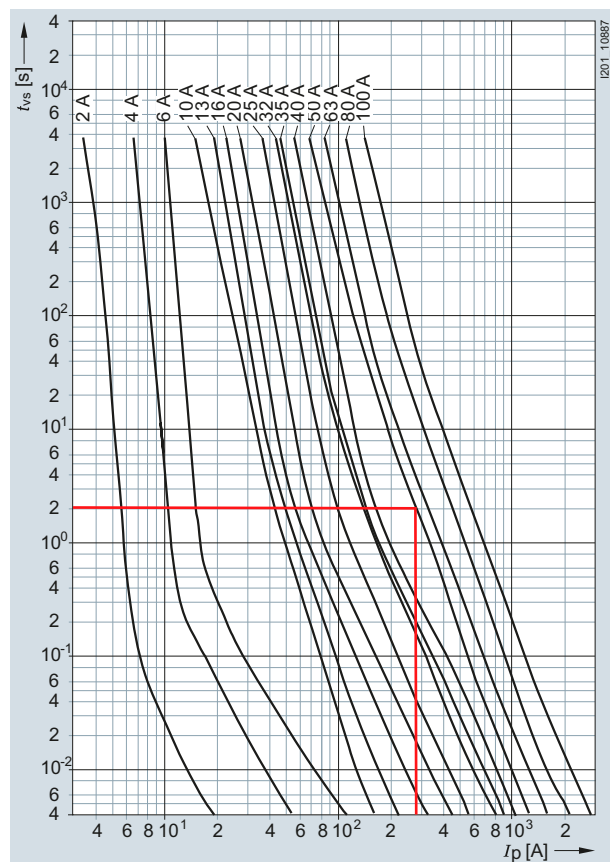
Characteristic curves

Series 5SE2

16 63 100

Sizes: D01, D02, D03
 Operational class: gG
 Rated voltage: 400 V AC/250 V DC
 Rated current: 2 ... 100 A

Time/current characteristics diagram



Melting I^2t values diagram

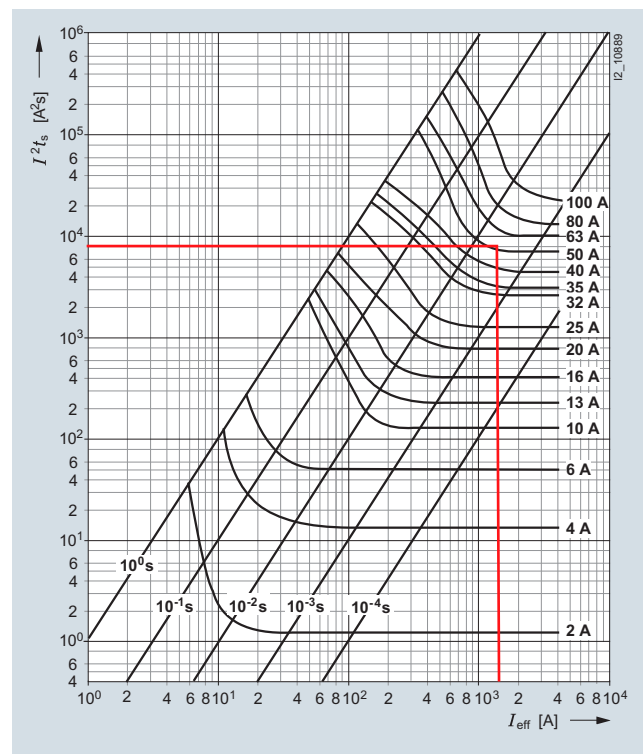
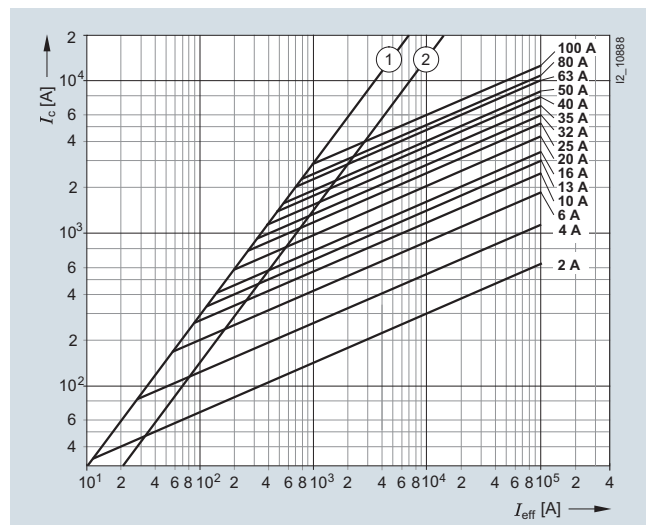


Table [see page 14](#).

Current limiting diagram



- ① Peak short-circuit current with largest DC component
- ② Peak short-circuit current without DC component

Fuse Systems

NEOZED Fuse Systems

NEOZED fuse links

Series 5SE2

Sizes: D01, D02, D03
 Operational class: gG
 Rated voltage: 400 V AC/250 V DC
 Rated current: 2 ... 100 A

Type	I_n	P_v	$\Delta\theta$	I^2t_s 1 ms	4 ms	I^2t_a 230 V AC ($t \leq 4$ ms)	400 V AC
	A	W	K	A ² s	A ² s	A ² s	A ² s
5SE2302	2	1.6	19	1.2	1.4	2.9	3.9
5SE2304	4	1.3	14	12.5	13.6	22	30
5SE2306	6	1.7	19	46.7	48	58	75
5SE2310	10	1.3	16	120	136	220	280
5SE2013-2A	13	2.0	23	220	244	290	370
5SE2316	16	2.1	24	375	410	675	890
5SE2320	20	2.4	26	740	810	1250	1650
5SE2325	25	3.2	33	1210	1300	1900	2600
5SE2332	32	3.6	34	2560	2800	4300	5500
5SE2335	35	3.8	36	3060	3500	5100	6500
5SE2340	40	4.0	37	4320	4800	7900	9500
5SE2350	50	4.2	38	6750	7400	10500	13000
5SE2363	63	5.3	45	10000	10900	16000	20500
5SE2280	80	5.3	43	13000	15400	25000	34500
5SE2300	100	6.4	47	22100	30000	46000	60000

Fuse Systems

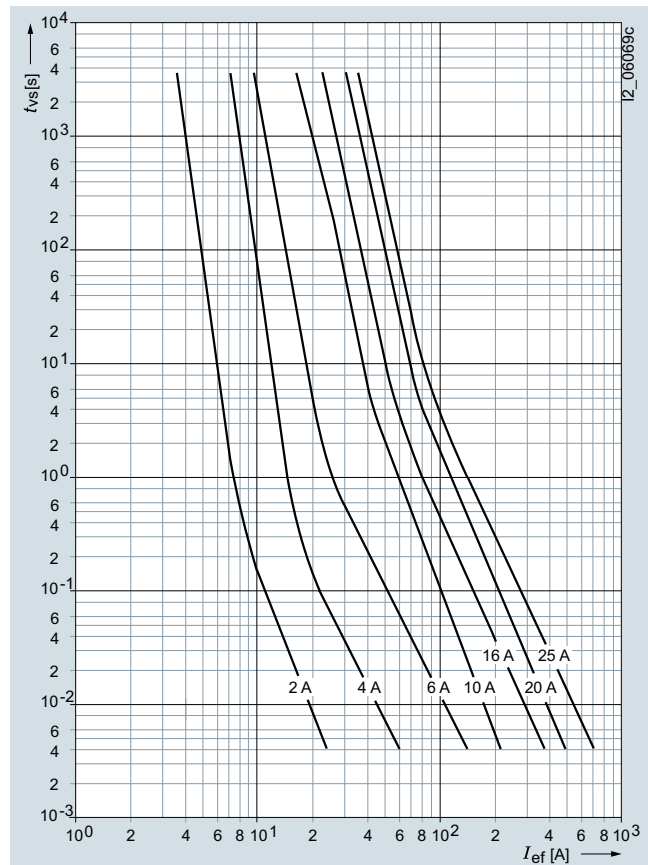
DIAZED fuse systems

Characteristic curves

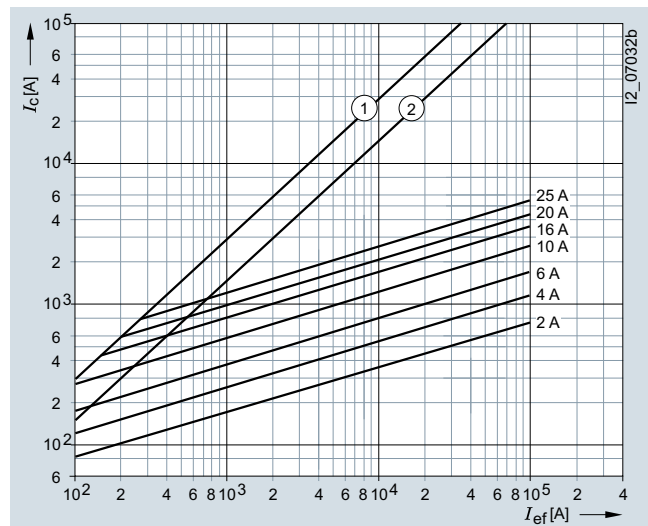
Series 5SA2

Size: E16
 Characteristics: Slow
 Rated voltage: 500 V AC/500 V DC
 Rated current: 2 ... 25 A

Time/current characteristics diagram

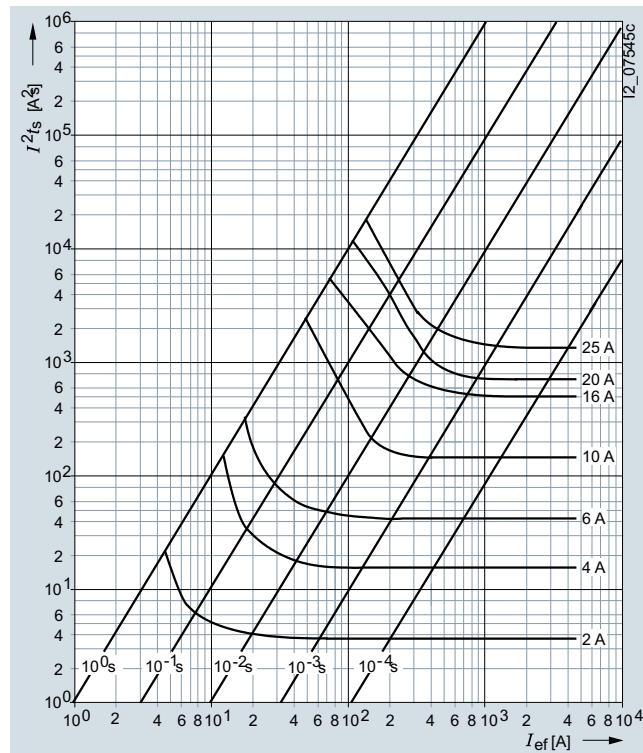


Current limiting diagram



- ① Peak short-circuit current with largest DC component
 ② Peak short-circuit current without DC component

Melting I^2t values diagram

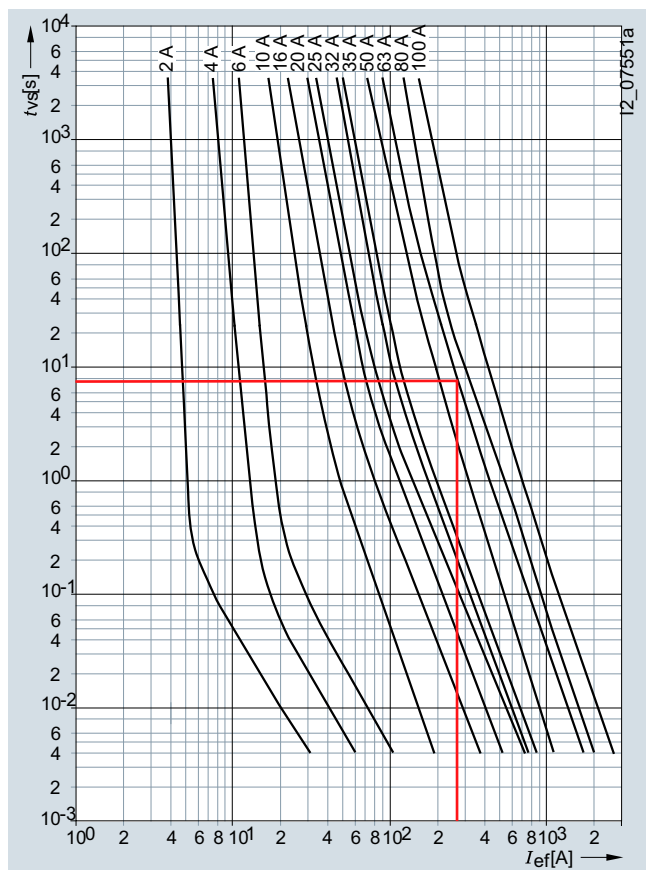
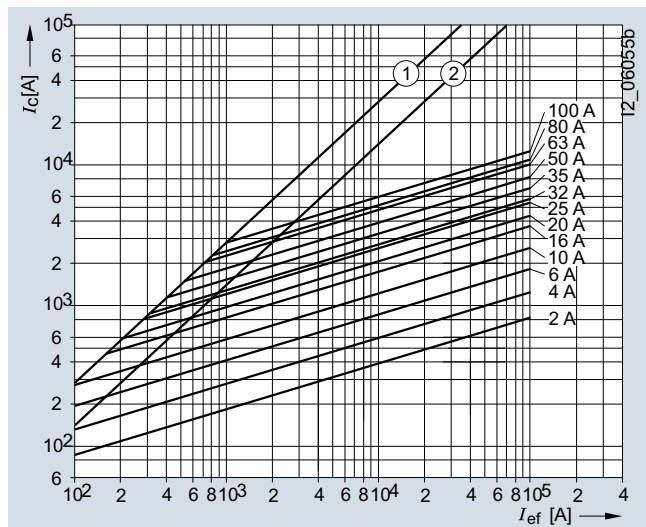


Type	I_n A	P_v W	$\Delta\theta$ K	I^2t_s 1 ms A ² s	4 ms A ² s
5SA211	2	0.85	15	1.2	2.3
5SA221	4	1.3	17	8.5	13
5SA231	6	1.9	14	40	80
5SA251	10	1.4	17	200	190
5SA261	16	2.4	30	290	550
5SA271	20	2.6	36	470	1990
5SA281	25	3.4	34	1000	2090

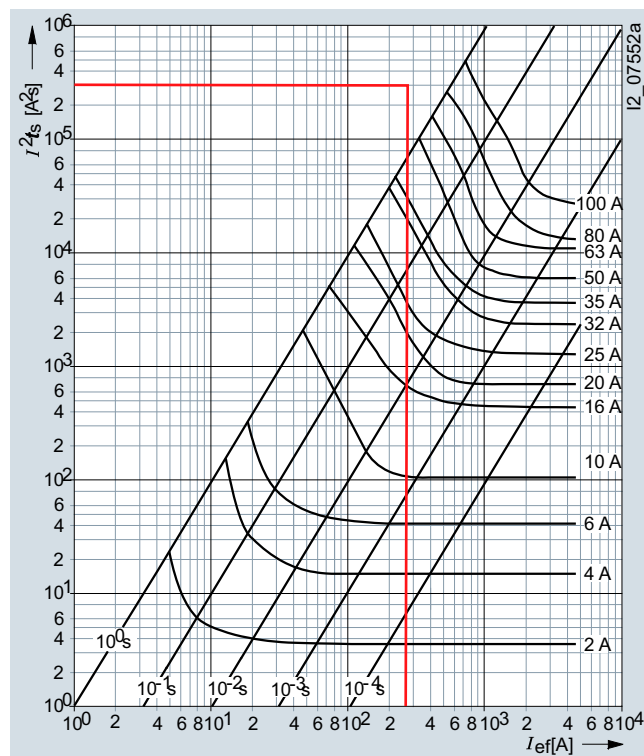
Type	I^2t_a 230 V AC A ² s	320 V AC A ² s	500 V AC A ² s
5SA211	6.6	7.8	0.7
5SA221	22	26	34
5SA231	66	76	100
5SA251	240	270	340
5SA261	890	950	1090
5SA271	1200	1350	1620
5SA281	2400	2600	3450

Series 5SB2, 5SB4, 5SC2

Size: DII, DIII, DIV
 Operational class: gG
 Rated voltage: 500 V AC/500 V DC
 Rated current: 2 ... 100 A

Time/current characteristics diagram**Current limiting diagram**

- ① Peak short-circuit current with largest DC component
 ② Peak short-circuit current without DC component

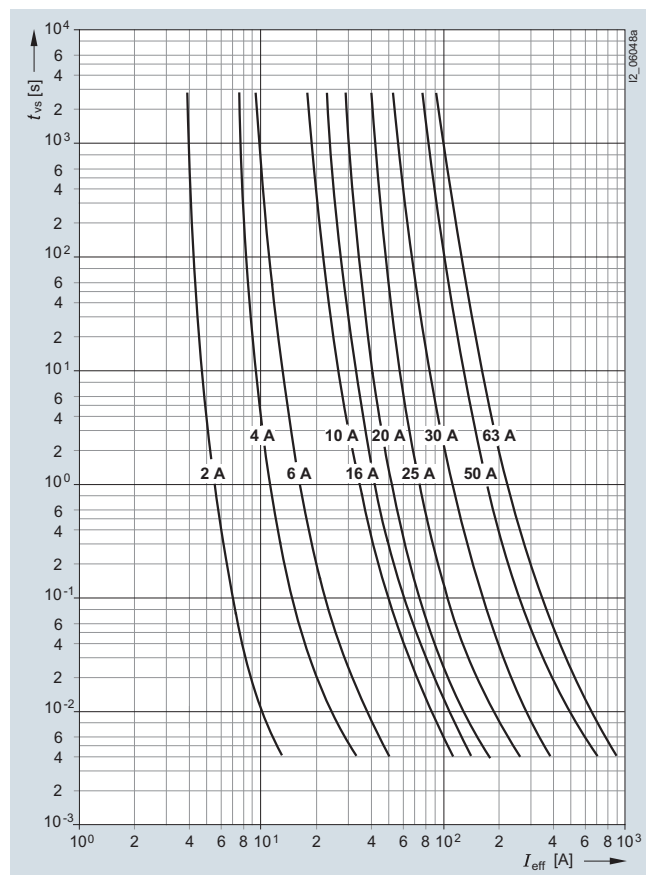
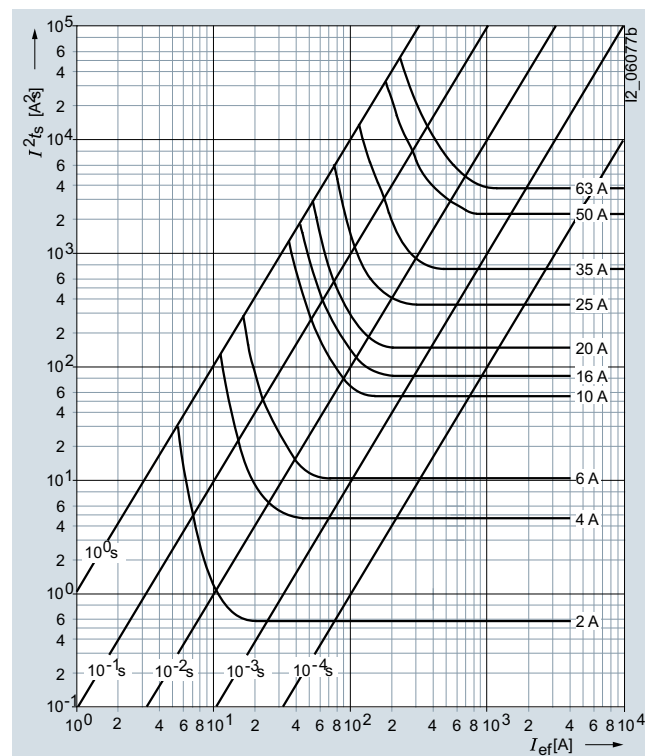
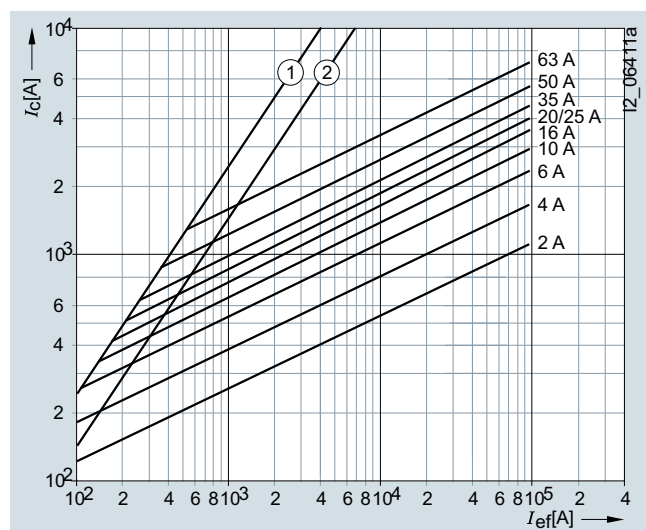
Melting I^2t values diagram

Type	I_n A	P_V W	$\Delta\theta$ K	I^2t_s 1 ms A ² s	I^2t_s 4 ms A ² s
5SB211	2	2.6	15	3.7	3.9
5SB221	4	2.0	13	15	16
5SB231	6	2.2	14	42	45
5SB251	10	1.6	20	120	140
5SB261	16	2.4	23	500	580
5SB271	20	2.6	26	750	1100
5SB281	25	3.4	38	1600	2000
5SB4010	32	3.6	23	2300	2500
5SB411	35	3.7	25	3450	3000
5SB421	50	5.7	41	6500	5200
5SB431	63	6.9	48	11000	12000
5SC211	80	7.5	33	14600	16400
5SC221	100	8.8	46	28600	30000

Type	I^2t_a 230 V AC A ² s	I^2t_a 320 V AC A ² s	I^2t_a 500 V AC A ² s
5SB211	6.6	8.8	10.7
5SB221	22	28	34
5SB231	66	85	100
5SB251	240	300	340
5SB261	890	1060	1090
5SB271	1200	1450	1620
5SB281	2400	3150	3450
5SB4010	3450	4150	4850
5SB411	5200	6200	7200
5SB421	9750	12350	14500
5SB431	16500	22200	26500
5SC211	23000	28500	32500
5SC221	44000	56000	65000

Series 5SD6

Size: DIII
 Operational class: Quick (railway network protection)
 Rated voltage: 750 V AC/750 V DC
 Rated current: 2 ... 63 A

Time/current characteristics diagram**Melting I^2t values diagram****Current limiting diagram**

- ① Peak short-circuit current with largest DC component
 ② Peak short-circuit current without DC component

Type	I_n A	P_v W	I^2t_s 4 ms A ² s	I^2t_a 500 V AC A ² s
5SD601	2	2.8	0.7	2
5SD602	4	4	4.5	13
5SD603	6	4.8	10	29
5SD604	10	4.8	50	135
5SD605	16	5.9	78	220
5SD606	20	6.3	125	380
5SD607	25	8.3	265	800
5SD608	35	13	550	1600
5SD610	50	16.5	1800	5500
5SD611	63	18	3100	9600

Fuse Systems

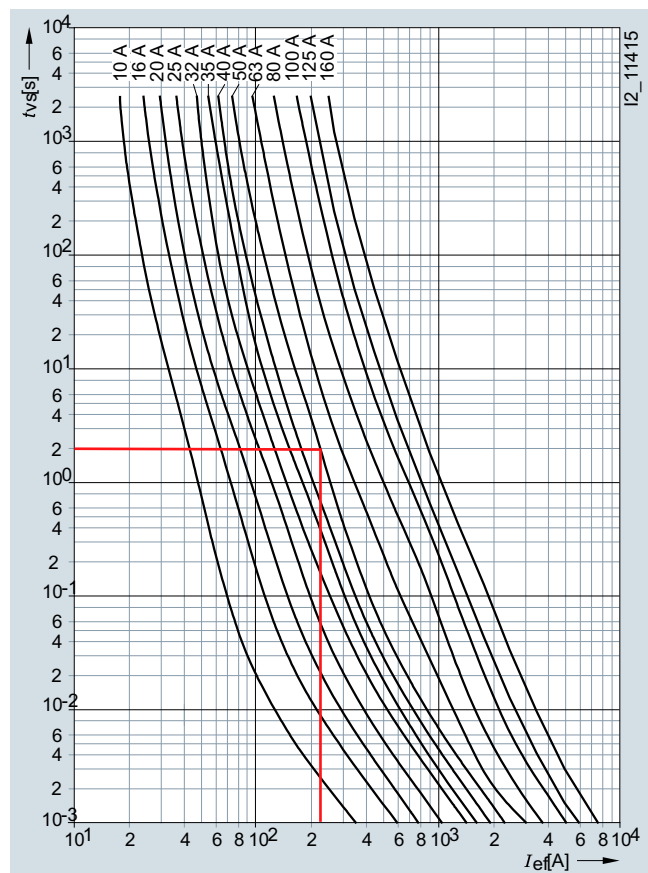
3NA, 3ND LV HRC Fuse Systems

LV HRC fuse links

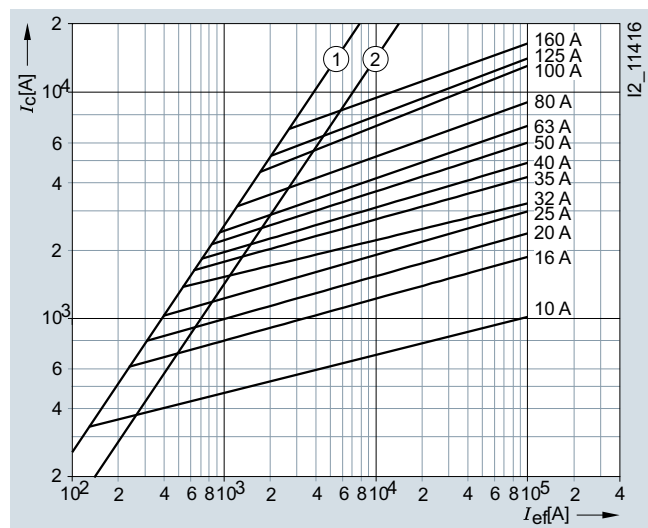
3NA68...-4/-4KK series

Size: 000, 00
 Operational class: gG
 Rated voltage: 400 V AC
 Rated current: 10 ... 160 A

Time/current characteristics diagram

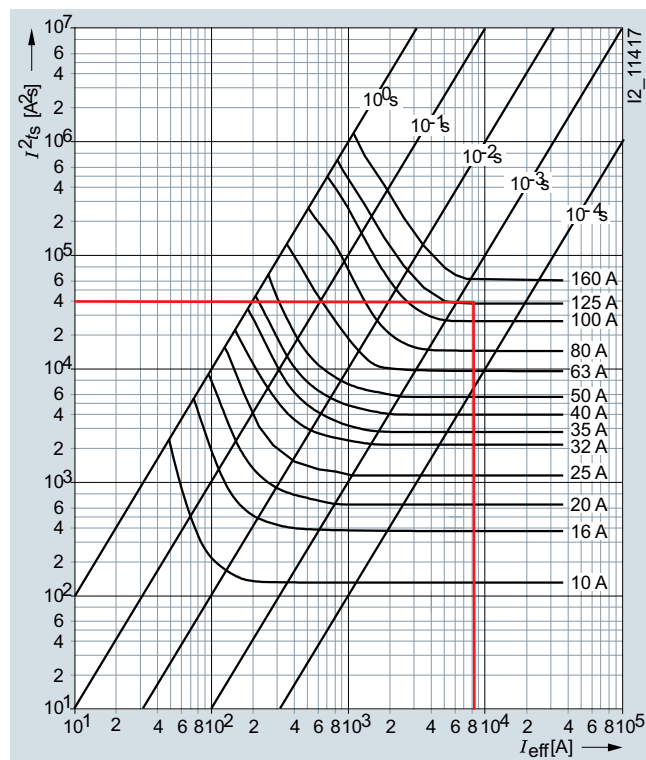


Current limiting diagram



- ① Peak short-circuit current with largest DC component
 ② Peak short-circuit current without DC component

Melting I^2t values diagram

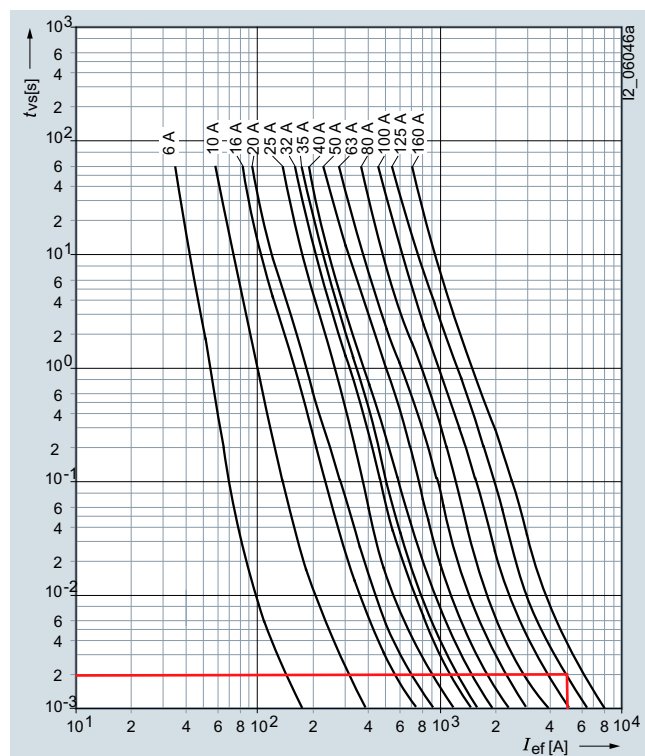
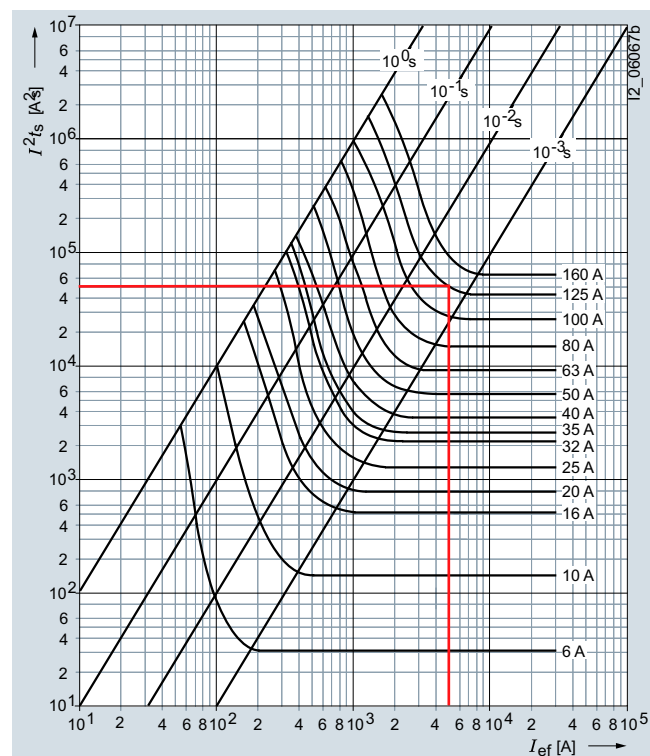
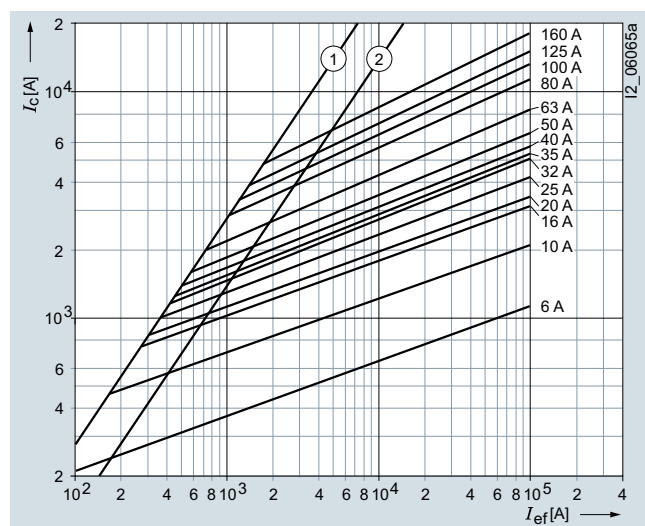


Type	I_n A	P_v W	$\Delta\theta$ K	I^2t_s 1 ms A ² s	4 ms A ² s
3NA6803-4	10	1.0	8	120	130
3NA6805-4	16	1.7	11	370	420
3NA6807-4	20	2.0	15	670	750
3NA6810-4	25	2.3	17	1200	1380
3NA6812-4	32	2.6	18	2200	2500
3NA6814-4	35	2.7	21	3000	3300
3NA6817-4	40	3.1	24	4000	4500
3NA6820-4	50	3.8	25	6000	6800
3NA6822-4	63	3.9	23	9300	10250
3NA6824-4, 3NA6824-4KK	80	4.9	26	14200	18300
3NA6830-4, 3NA6830-4KK	100	5.4	29	25600	33600
3NA6832-4	125	8.9	44	36000	50000
3NA6836-4	160	11.3	52	58000	85000

Type	I^2t_a 230 V AC A ² s	400 V AC A ² s
3NA6803-4	180	265
3NA6805-4	580	750
3NA6807-4	1000	1370
3NA6810-4	1800	2340
3NA6812-4	3400	4550
3NA6814-4	4900	6750
3NA6817-4	6100	8700
3NA6820-4	9100	11600
3NA6822-4	12400	17900
3NA6824-4, 3NA6824-4KK	27000	38000
3NA6830-4, 3NA6830-4KK	48300	69200
3NA6832-4	70000	91300
3NA6836-4	120000	158000

3ND18 series

Size: 000, 00
 Operational class: aM
 Rated voltage: 500 V AC
 Rated current: 6 ... 160 A

Time/current characteristics diagram**Melting I^2t values diagram****Current limiting diagram**

- ① Peak short-circuit current with largest DC component
 ② Peak short-circuit current without DC component

Type	I_n	P_v	$\Delta\theta$	I^2t_s	
	A	W	K	1 ms A ² s	4 ms A ² s
3ND1801	6	0.8	7	32	55
3ND1803	10	0.5	5	150	260
3ND1805	16	0.8	7	570	800
3ND1807	20	1	8	830	1200
3ND1810	25	1.2	9	1400	2000
3ND1812	32	1.5	10	2300	3300
3ND1814	35	1.8	11	2600	3800
3ND1817	40	2	12	3700	5500
3ND1820	50	2.4	14	5800	8400
3ND1822	63	3.3	17	9300	13000
3ND1824	80	4.5	20	15000	21000
3ND1830, 3ND1830-8	100	4.9	18	26000	37000
3ND1832	125	6.3	22	41000	60000
3ND1836	160	9.3	31	64000	92000

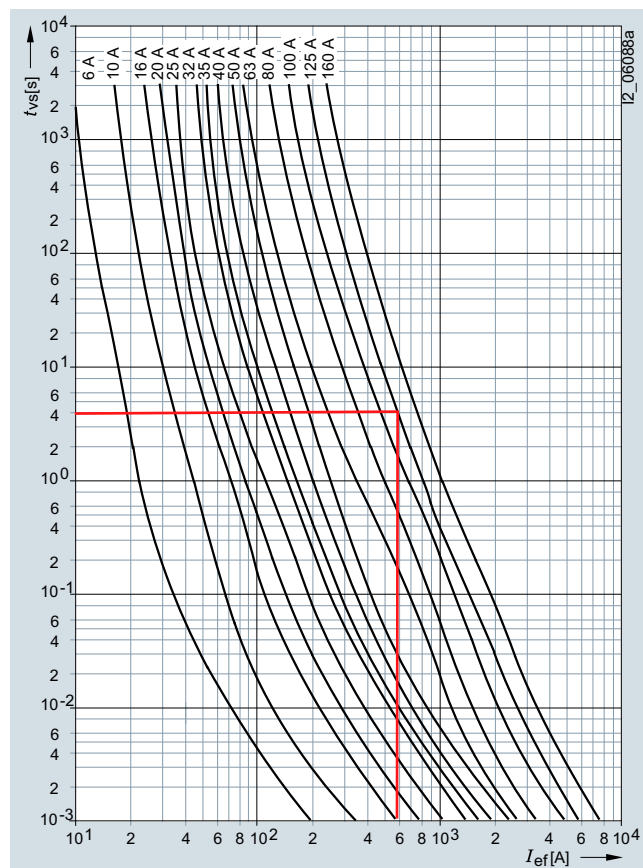
Type	I^2t_a		
	230 V AC A ² s	400 V AC A ² s	500 V AC A ² s
3ND1801	60	75	110
3ND1803	280	320	430
3ND1805	1000	1300	1600
3ND1807	1300	1600	2200
3ND1810	2200	2800	3300
3ND1812	3800	4500	5400
3ND1814	4200	5100	6300
3ND1817	5700	7200	9300
3ND1820	5200	10500	12500
3ND1822	15000	16500	21000
3ND1824	21500	27000	34000
3ND1830, 3ND1830-8	44000	56000	76000
3ND1832	76000	98000	135000
3ND1836	105000	130000	170000

Characteristic curves

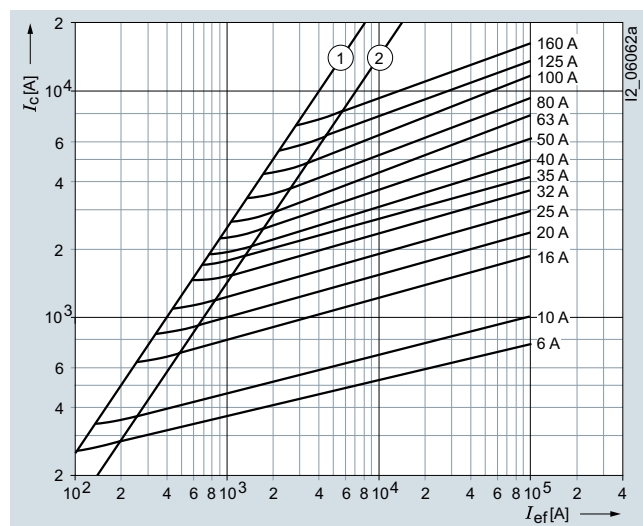
3NA30 series

Size: 0
Operational class: gG
Rated voltage: 500 V AC/440 V DC
Rated current: 6 ... 160 A

Time/current characteristics diagram

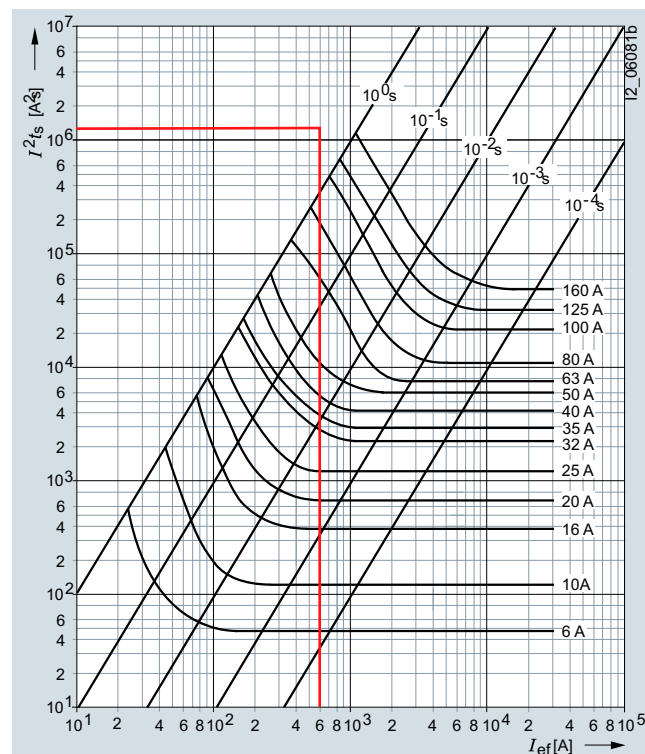


Current limiting diagram



- ① Peak short-circuit current with largest DC component
② Peak short-circuit current without DC component

Melting I^2t values diagram



Type	I_n A	P_v W	$\Delta\theta$ K	I^2t_s 1 ms A ² s	4 ms A ² s
3NA3001	6	1.5	6	46	50
3NA3003	10	1	9	120	130
3NA3005	16	1.9	11	370	420
3NA3007	20	2.3	13	670	750
3NA3010	25	2.7	15	1200	1380
3NA3012	32	3	13	2200	2400
3NA3014	35	3	17	3000	3300
3NA3017	40	3.4	17	4000	4500
3NA3020	50	4.5	24	6000	6800
3NA3022	63	5.8	27	7700	9800
3NA3024	80	7	34	12000	16000
3NA3030	100	8.2	37	24000	30600
3NA3032	125	10.2	38	36000	50000
3NA3036	160	13.5	44	58000	85000

Type	I^2t_a 230 V AC A ² s	400 V AC A ² s	500 V AC A ² s
3NA3001	80	110	150
3NA3003	180	265	370
3NA3005	580	750	1000
3NA3007	1000	1370	1900
3NA3010	1800	2340	3300
3NA3012	3400	4550	6400
3NA3014	4900	6750	9300
3NA3017	6100	8700	12100
3NA3020	9100	11600	16000
3NA3022	14200	19000	26500
3NA3024	23100	30700	43000
3NA3030	40800	56200	80000
3NA3032	70000	91300	130000
3NA3036	120000	158000	223000

Fuse Systems

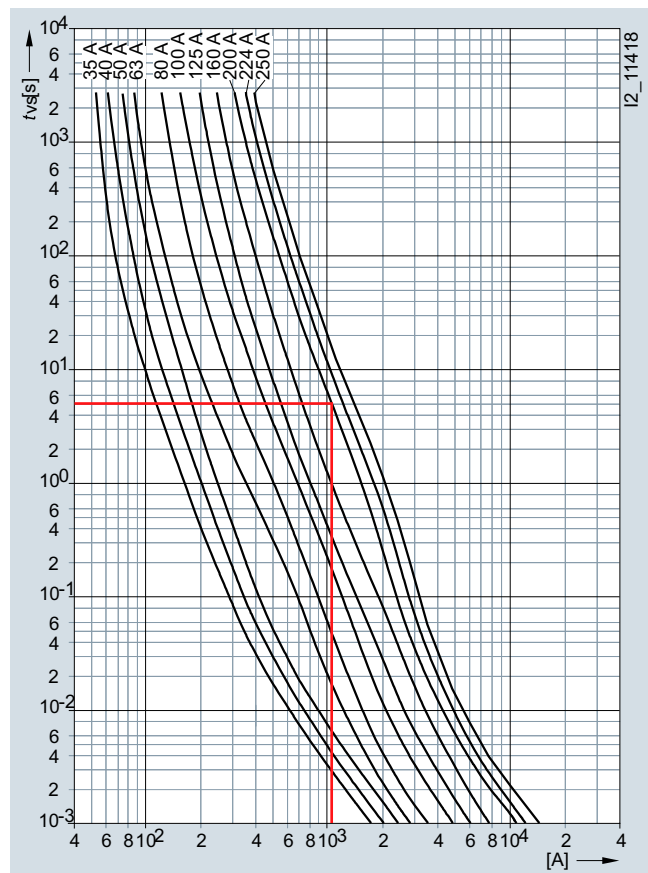
3NA, 3ND LV HRC Fuse Systems

LV HRC fuse links

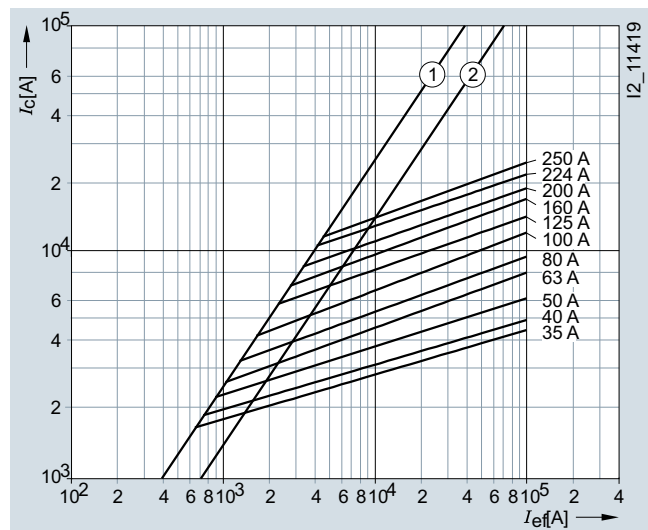
3NA61...-4 series

Size: 1
Operational class: gG
Rated voltage: 400 V AC
Rated current: 35 ... 250 A

Time/current characteristics diagram

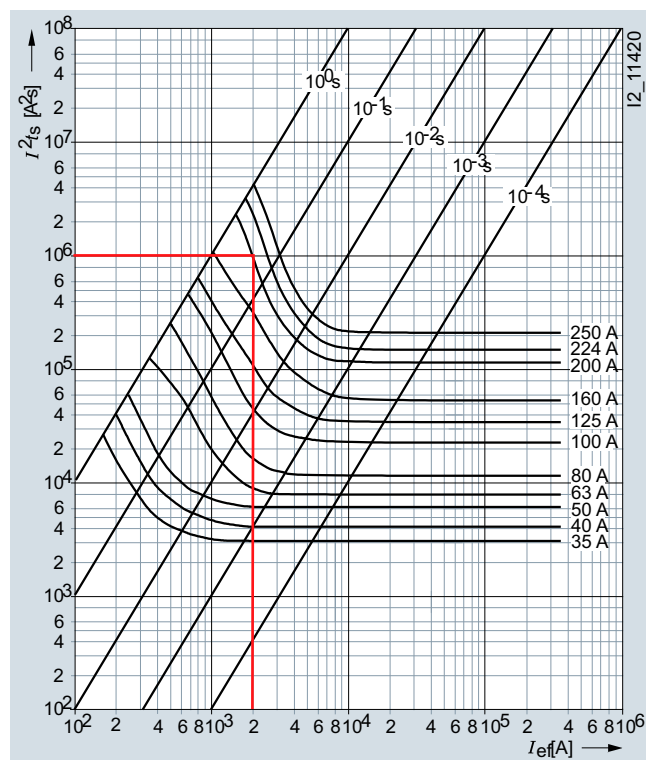


Current limiting diagram



- ① Peak short-circuit current with largest DC component
② Peak short-circuit current without DC component

Melting I^2t values diagram



Type	I_n A	P_v W	$\Delta\theta$ K
3NA6114-4	35	3.2	16
3NA6117-4	40	3.6	16
3NA6120-4	50	4.6	20
3NA6122-4	63	6.0	21
3NA6124-4	80	7.5	29
3NA6130-4	100	8.9	30
3NA6132-4	125	10.7	31
3NA6136-4	160	13.9	34
3NA6140-4	200	15.0	36
3NA6142-4	224	16.1	37
3NA6144-4	250	17.3	39

Type	I^2t_s 1 ms A ² s	I^2t_s 4 ms A ² s	I^2t_a 230 V AC A ² s	I^2t_a 400 V AC A ² s
3NA6114-4	3000	3300	4900	6750
3NA6117-4	4000	4500	6100	8700
3NA6120-4	6000	6800	9100	11600
3NA6122-4	7700	9800	14200	19000
3NA6124-4	12000	16000	23100	30700
3NA6130-4	24000	30600	40800	56200
3NA6132-4	36000	50000	70000	91300
3NA6136-4	58000	85000	120000	158000
3NA6140-4	115000	135000	218000	285000
3NA6142-4	145000	170000	299000	392000
3NA6144-4	205000	230000	420000	551000

Fuse Systems

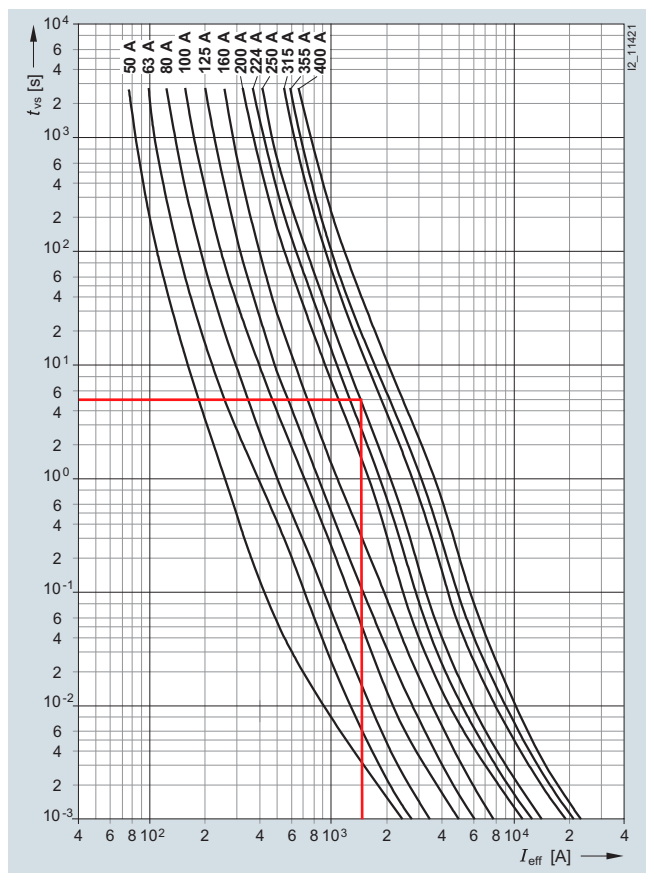
3NA, 3ND LV HRC Fuse Systems

LV HRC fuse links

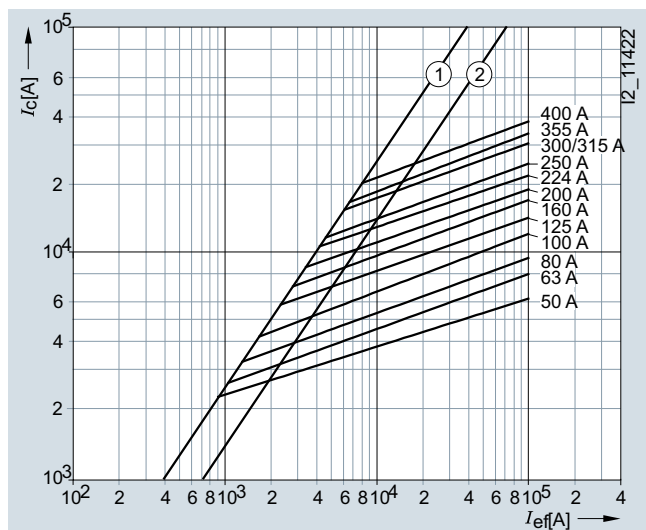
3NA62...-4 series

Size: 2
Operational class: gG
Rated voltage: 400 V AC
Rated current: 50 ... 400 A

Time/current characteristics diagram

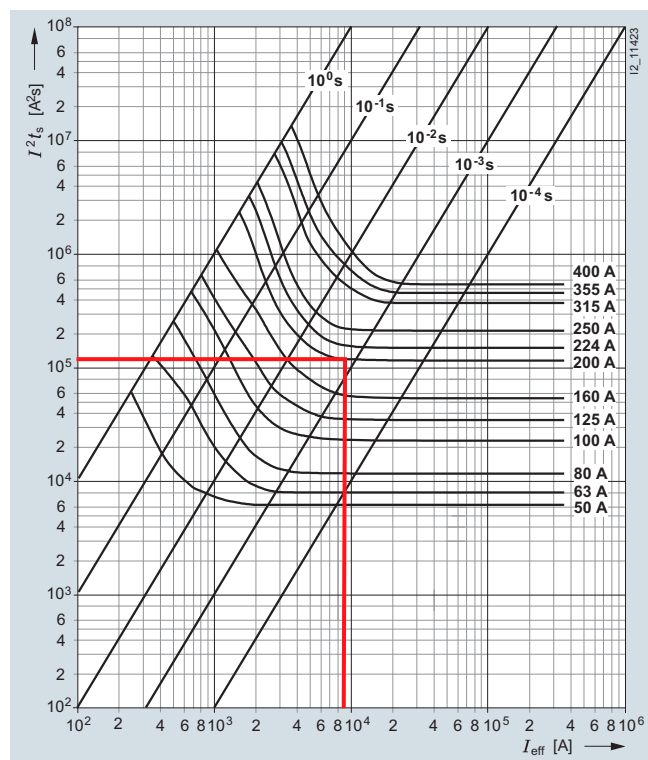


Current limiting diagram



- ① Peak short-circuit current with largest DC component
- ② Peak short-circuit current without DC component

Melting I^2t values diagram



Type	I_n A	P_v W	$\Delta\theta$ K
3NA6220-4	50	4.7	16
3NA6222-4	63	5.9	16
3NA6224-4	80	6.8	21
3NA6230-4	100	7.4	22
3NA6232-4	125	9.8	27
3NA6236-4	160	12.6	34
3NA6240-4	200	14.9	33
3NA6242-4	224	15.4	31
3NA6244-4	250	17.9	38
3NA6250-4	300	19.4	34
3NA6252-4	315	21.4	35
3NA6254-4	355	26.0	49
3NA6260-4	400	27.5	52

Type	I^2t_s 1 ms A^2s	I^2t_s 4 ms A^2s	I^2t_a 230 V AC A^2s	I^2t_a 400 V AC A^2s
3NA6220-4	6000	6800	9100	11600
3NA6222-4	7700	9800	14200	19000
3NA6224-4	12000	16000	23100	30700
3NA6230-4	24000	30600	40800	56200
3NA6232-4	36000	50000	70000	91300
3NA6236-4	58000	85000	120000	158000
3NA6240-4	115000	135000	218000	285000
3NA6242-4	145000	170000	299000	392000
3NA6244-4	205000	230000	420000	551000
3NA6250-4	361000	433000	670000	901000
3NA6252-4	361000	433000	670000	901000
3NA6254-4	441000	538000	800000	1060000
3NA6260-4	529000	676000	1155000	1515000

Fuse Systems

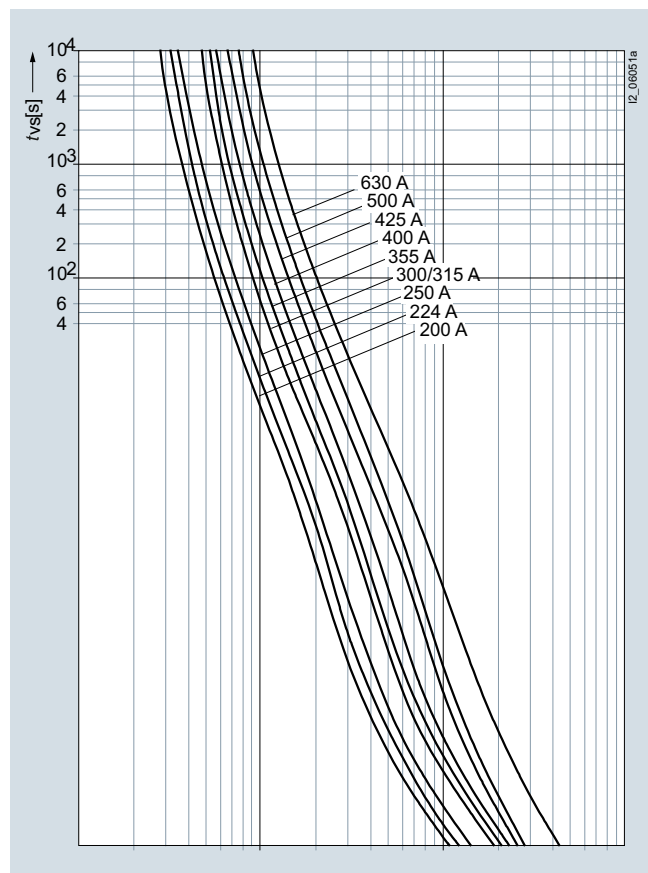
3NA, 3ND LV HRC Fuse Systems

LV HRC fuse links

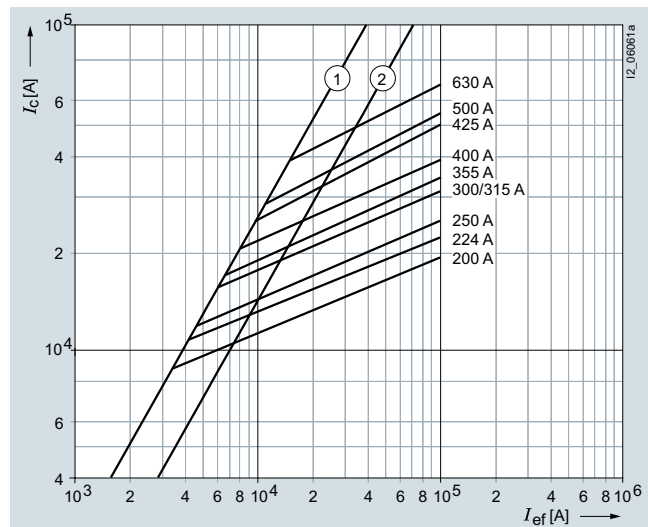
3NA33 series

Size: 3
Operational class: gG
Rated voltage: 500 V AC/440 V DC
Rated current: 200 ... 630 A

Time/current characteristics diagram

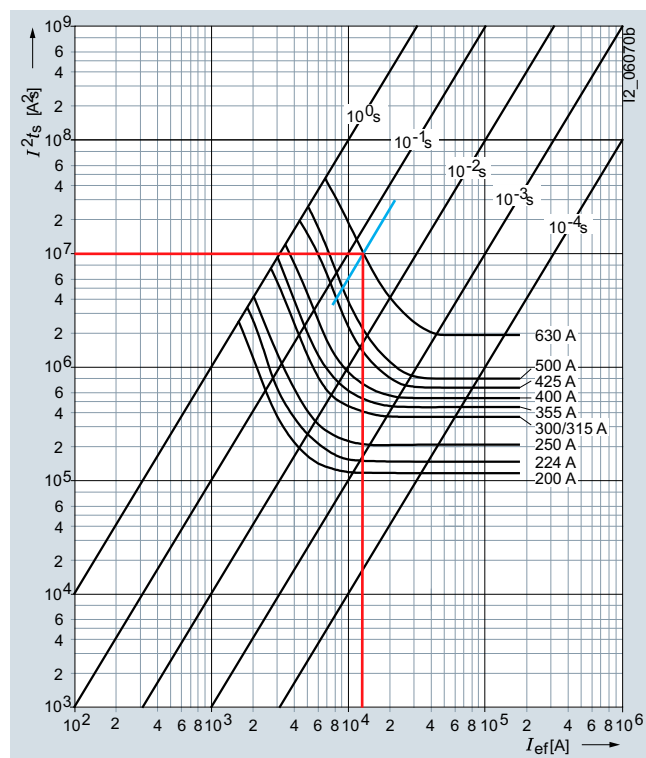


Current limiting diagram



- ① Peak short-circuit current with largest DC component
② Peak short-circuit current without DC component

Melting I^2t values diagram



Type	I_n A	P_v W	$\Delta\theta$ K	I^2t_s 1 ms A ² s	I^2t_s 4 ms A ² s
3NA3340	200	14.9	32	115000	135000
3NA3342	224	15.4	31	145000	170000
3NA3344	250	17.9	36	205000	230000
3NA3350	300	19.4	19	361000	433000
3NA3352	315	21.4	22	361000	433000
3NA3354	355	26.0	26	441000	538000
3NA3360	400	27.5	28	529000	676000
3NA3362	425	26.5	34	650000	970000
3NA3365	500	36.5	41	785000	1270000
3NA3372	630	44.0	50	1900000	2700000

Type	I^2t_a 230 V AC A ² s	I^2t_a 400 V AC A ² s	I^2t_a 500 V AC A ² s
3NA3340	218000	285000	400000
3NA3342	299000	392000	550000
3NA3344	420000	551000	780000
3NA3350	670000	901000	1275000
3NA3352	670000	901000	1275000
3NA3354	800000	1060000	1500000
3NA3360	1155000	1515000	2150000
3NA3362	1515000	1856000	2270000
3NA3365	1915000	2260000	2700000
3NA3372	3630000	4340000	5400000

Fuse Systems

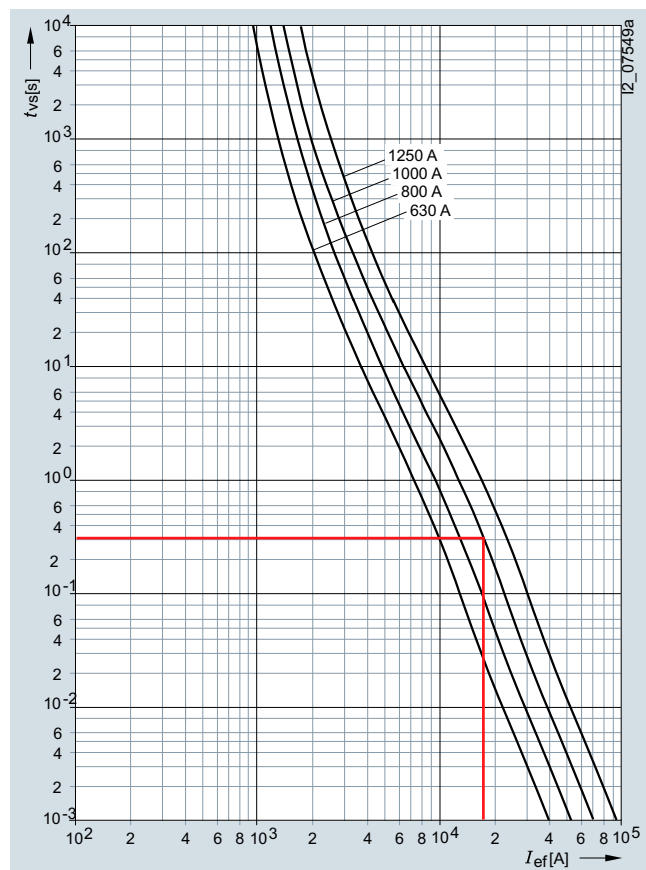
3NA, 3ND LV HRC Fuse Systems

LV HRC fuse links

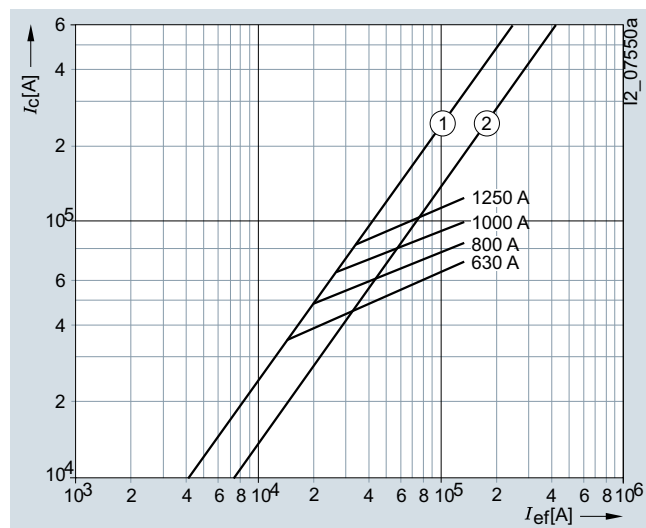
3NA34 series

Size: 4 (IEC design)
 Operational class: gG
 Rated voltage: 500 V AC/440 V DC
 Rated current: 630 ... 1250 A

Time/current characteristics diagram

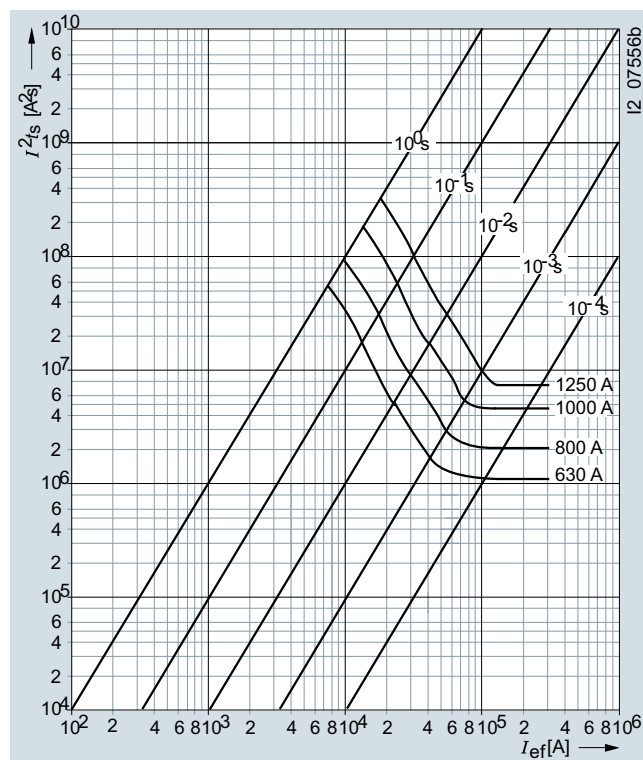


Current limiting diagram



- ① Peak short-circuit current with largest DC component
 ② Peak short-circuit current without DC component

Melting I^2t values diagram



Type	I_n A	P_v W	$\Delta\theta$ K	I^2t_s 1 ms A ² s	I^2t_s 4 ms A ² s
3NA3472	630	47	37	1900000	2700000
3NA3475	800	59	43	3480000	5620000
3NA3480	1000	74	56	7920000	10400000
3NA3482	1250	99	65	11880000	18200000

Type	I^2t_a 230 V AC A ² s	I^2t_a 400 V AC A ² s	I^2t_a 500 V AC A ² s
3NA3472	3630000	4340000	5400000
3NA3475	7210000	8510000	10400000
3NA3480	13600000	16200000	19000000
3NA3482	23900000	29100000	34800000